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CLIMATE &amp; SUSTAINABILITY

# The Kiln Runs Backwards

*Making cement drives a reaction that the open air spends the next century quietly undoing, at a rate set less by chemistry than by the thickness of a wall.*

2,574 words · 14-minute read

Two hundred years ago last autumn, a bricklayer in Leeds named Joseph Aspdin took out a patent on a way of making artificial stone by burning a mixture of limestone and clay and grinding the result to powder [1]. He called it Portland cement, after the pale oolitic limestone quarried on the Dorset coast, because he wanted builders to believe the stuff would set into something like rock. The name was marketing. The chemistry turned out to be more literal than he could have known: the powder really does want to become limestone again, and given a few decades and a supply of ordinary air, a great deal of it does.

That return has recently become large enough to appear in the world's carbon accounts. The manufacture of cement is responsible for something over seven per cent of anthropogenic carbon dioxide, and roughly three-fifths of that is not combustion at all but the chemical decomposition of limestone — an emission no change of fuel can touch [2]. Set against it, running quietly in every wall, pavement and rubble pile ever built, is a reverse reaction that the Global Carbon Budget now carries as a standing line item of about 0.2 billion tonnes of carbon a year [3]. What follows is an account of how large that return is, why it takes so long, and why the credit for most of it belongs not to concrete but to the thin grey layer holding bricks together.

## Two Tonnes of Stone, One of Powder

The reaction at the heart of a cement kiln is the simplest in heavy industry. Calcium carbonate, heated hard enough, gives up its carbon dioxide and leaves quicklime behind:



Everything downstream follows from the molar masses. Calcium carbonate is 56.03 per cent lime and 43.97 per cent carbon dioxide by weight, so every kilogram of lime costs 0.785 kg of carbon dioxide, released as a matter of stoichiometry rather than of efficiency. Clinker — the fused nodules that come out of the kiln and are ground into cement — is about 65 per cent lime, and 0.65 times 0.785 gives 0.510 tonnes of carbon dioxide per tonne of clinker. That is the number national inventory compilers use [4]. Indeed the two default emission factors in circulation, 0.5070 and 0.5101, are not competing measurements: they are the same calculation run on lime contents of 64.6 and 65.0 per cent, and both

reproduce to four decimal places from the molar masses alone. Cement is not pure clinker — the world average is roughly 72 per cent, the rest being slag, fly ash and ground limestone [18] — so the process burden per tonne of cement sold is nearer 0.37 tonnes.

The energy tells a similar story. A tonne of clinker needs 1.16 tonnes of carbonate decomposed, which is 11,591 moles, which at 179 kJ each is 2.08 gigajoules of pure chemistry. Some of it comes back: the lime then combines with silica to form alite, the phase that gives cement its strength, and that combination is exothermic to the tune of about 115 kJ for every mole formed, or roughly 0.30 GJ per tonne of clinker at ordinary alite contents. Net, the irreducible heat of making clinker is about 1.77 GJ per tonne, which is where the published theoretical minimum of roughly 1.75 GJ per tonne comes from [5]. Real kilns burn something closer to 3.5. The gap is heat loss, not chemistry, and it is the part of the problem that engineering can attack.

## A Reaction the Air Wants to Undo

Calcination is violently unfavourable at room temperature, which is precisely why it needs a kiln. Run the standard thermodynamic numbers the other way and the magnitude is startling. With an enthalpy of 179 kJ per mole and an entropy change of 160.2 J per mole per kelvin, the free energy of decomposition at 20 degrees Celsius is 132 kJ per mole, and the equilibrium partial pressure of carbon dioxide over calcium carbonate is about 3 times 10 to the minus 24 bar.

The atmosphere currently sits at 422 parts per million, or 4.3 times 10 to the minus 4 bar. The air outside every building on earth is supersaturated with respect to limestone by twenty orders of magnitude. Thermodynamically there is no question whatever about which way the reaction runs; a set of concrete steps is not in equilibrium with the air around it and never will be until every available calcium atom has gone back to carbonate.

### THE OBVIOUS QUESTION

If the driving force is that overwhelming, why is there any lime left in a hundred-year-old bridge at all?

Because nothing in this story is limited by thermodynamics. In hardened cement the calcium sits as portlandite and as calcium silicate hydrate inside a pore network saturated with alkaline solution, and carbon dioxide has to dissolve into that solution and diffuse inward before it can react at all. The reaction itself is fast. The delivery is not — and the delivery is the whole subject.

## Why a Century Is Not Long Enough

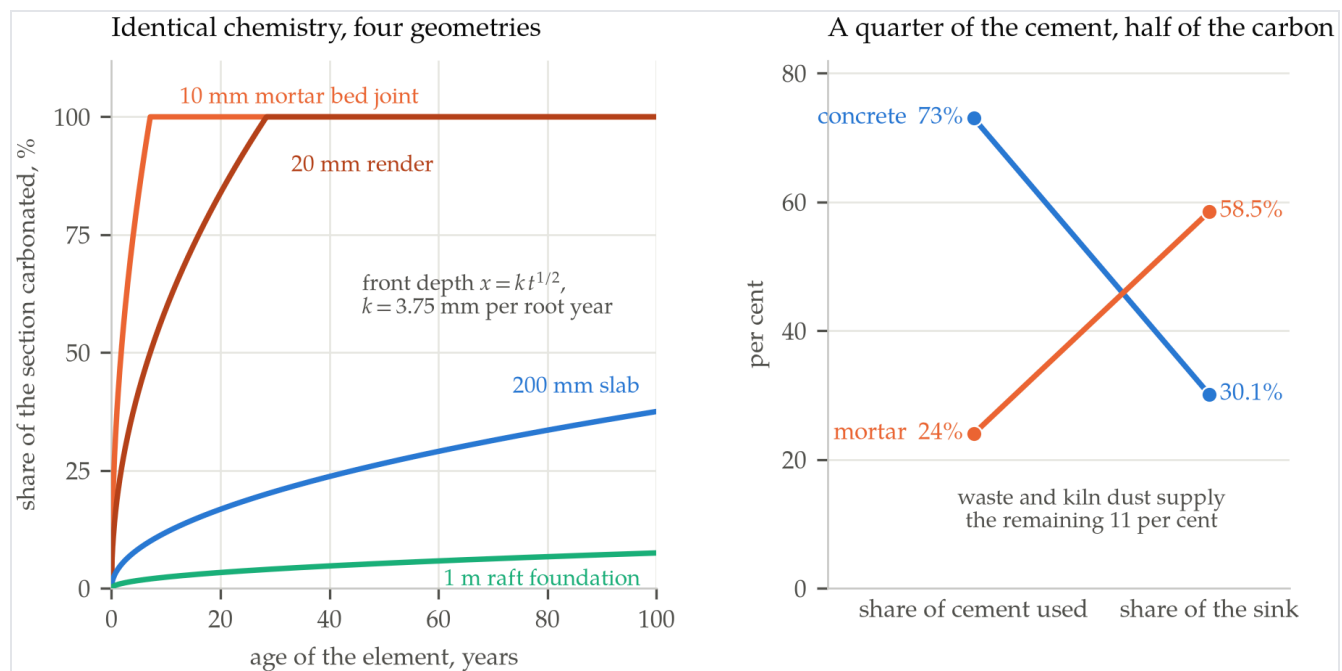
Treat the carbonated skin as a diffusion barrier and the mathematics is the same sharp-front problem that governs oxidation scales and case hardening. Carbon dioxide diffuses through the already-reacted outer layer at an effective diffusivity  $D_e$ , meets unreacted material at depth  $x$ , and is consumed there. If

the concrete binds  $a$  moles of carbon dioxide per cubic metre and the external concentration is  $C_s$ , conservation across the advancing front gives

$$x(t) = \sqrt{\frac{2D_e C_s}{a} t} = k\sqrt{t}$$

which is the square-root law every durability code is built on [6], [7]. Put ordinary numbers in it. At 422 parts per million and 20 degrees Celsius the air holds 0.0175 moles of carbon dioxide per cubic metre. A concrete with 300 kg of cement per cubic metre at 65 per cent lime binds about 3,480 moles per cubic metre — roughly two hundred thousand times the concentration in the air feeding it, which is the real reason the front crawls. With an effective diffusivity of 5 times 10 to the minus 8 square metres per second, typical of carbonated structural concrete, the coefficient  $k$  works out at 4.0 mm per square root of a year; at one-fifth that diffusivity it is 1.8. Field surveys of in-service structures report coefficients across very much the same band, from about 2 for dense concrete in sheltered conditions to 6 and above in cyclically wet exposure and into the teens for poor material [7]. A first-principles calculation with two textbook inputs lands where the measurements are.

The consequence is a reaction that never finishes and never stops. Doubling the depth costs four times the wait. Whatever the front has achieved in ten years takes a further thirty to double, and the last millimetre of a thick section is effectively unreachable.



**FIGURE** Four elements made of the same material, carbonating at wildly different rates, and what that does to the global accounts. Left: the fraction of a section converted, under a front advancing as  $k$  times the root of elapsed time with  $k$  of 3.75 mm per root year. Right: shares of cement use, from the 1928-2024 account, against shares of the carbonation sink, from the 1930-2021 account.

## A Quarter of the Cement, Half of the Carbon

Now put that law next to the shapes cement actually gets used in, and something counterintuitive falls out. A 10 mm mortar bed joint between two bricks is carbonated through in seven years. A 20 mm render coat is finished in twenty-eight. A 200 mm floor slab attacked from both faces would need seven centuries, and a metre-thick raft foundation nearly eighteen thousand years. The chemistry is identical in all four. Only the distance differs, and the distance is squared.

So the global sink is not where the cement is. Concrete takes roughly 73 per cent of the world's cement and supplies about 30 per cent of the carbonation sink; mortar and render take roughly 24 per cent and supply between 48 and 59 per cent, depending on which account and which period one takes [8], [9]. Per tonne of cement, in other words, mortar has absorbed something between three and a half and six times as much carbon dioxide as concrete — a spread worth printing in full, because the two leading research groups do not agree on it. Structural engineering gets the credit for concrete; the carbon sink is largely the work of plasterers.

The same logic explains the other half of the life cycle. Demolition does not end the process, it accelerates it: crushing a slab into aggregate multiplies the exposed surface by orders of magnitude and exposes cores that had been sealed for decades [10]. A structure's largest single day of carbon uptake is usually the day it is destroyed.

## Twenty-One Billion Tonnes, and Counting

Five research accounts now exist of how much has come back, and they are close enough to be credible and far enough apart to be honest about.

| Account               | Window    | Uptake, Gt CO <sub>2</sub> | Share of process emissions |
|-----------------------|-----------|----------------------------|----------------------------|
| Xi et al. 2016 [11]   | 1930-2013 | 16.5                       | 43 %                       |
| Guo et al. 2021 [12]  | 1930-2019 | 21.0                       | —                          |
| Huang et al. 2023 [8] | 1930-2021 | 22.9                       | 55.1 %                     |
| Wu et al. 2024 [13]   | 1930-2023 | 23.9                       | 52.3 %                     |
| Niu et al. 2025 [9]   | 1928-2024 | 21.3                       | 46.1 %                     |

Two of these are worth checking rather than trusting. Xi and colleagues report their result as 4.5 billion tonnes of carbon, which converts to 16.5 billion tonnes of carbon dioxide at the ratio of molar masses. Huang and colleagues report 22.9 billion tonnes absorbed against 41.6 billion emitted and an offset of 55.1 per cent; the division gives 55.05, so all three of their headline figures confirm each other. The Global Carbon Budget's sink of 0.2 billion tonnes of carbon a year is the same quantity in different units as the "over 700 million tonnes of carbon dioxide" quoted elsewhere, since the conversion factor is 3.664 [3].

The current rate is 0.84 to 0.93 billion tonnes of carbon dioxide a year [9], [13], against process emissions of roughly 1.6 billion — so the world's standing stock of cement is now reabsorbing more than half of each year's calcination as it happens.

The spread between 43 and 55 per cent is not noise to be averaged away. It reflects genuine disagreement about service lives, demolition rates and how much of a crushed stockpile ever sees air, and the two most recent accounts — from overlapping author groups, on nearly the same window — differ by six percentage points. Anyone quoting a single number for this sink is quoting a modelling choice.

What can be stated exactly is the ceiling. Carbonation reverses calcination mole for mole: one molecule of carbon dioxide out of the kiln, one molecule back into the wall. Complete carbonation of every calcium atom that was ever a carbonate would therefore return 100 per cent of the process emissions and not one gram of the fuel emissions, which are roughly two-fifths of the total and are gone for good. A sink running at half its ceiling after a century is doing well, and can never do better than three-fifths of the industry's footprint however long one waits.

There is also a reason the cumulative share understates what is coming. Cement output doubled from two to four billion tonnes between 2003 and 2013, a continuously compounded 6.9 per cent a year, and under exponential growth exactly half of everything ever produced dates from within the last doubling time. Most of the world's cement is young, and young concrete has carbonated a centimetre at most. The stock has been chasing a flow that only stopped growing about a decade ago.

## What the Return Costs

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The sink is not free, and the bill is paid by the steel. Fresh concrete protects reinforcement not by sealing it but by surrounding it with pore water at pH 12 to 13, in which iron holds a passive oxide film a few nanometres thick. Carbonation consumes exactly the alkalinity that maintains that film. Behind the front the pore solution falls to about 9 or below, the film destabilises, and corrosion begins wherever moisture and oxygen are available [14], [15].

This is why reinforced concrete is designed with cover, and cover is nothing but a deliberately imposed diffusion delay. Eurocode 2 asks for 25 mm of durability cover in moderate-humidity exposure for a fifty-year design life, plus a construction tolerance [16]. At a coefficient of 3.75 mm per root year, 25 mm of cover is reached in 44 years; at 3.0 it is 69 years, and at 6.0 it is 17. The code's requirement and the square-root law agree closely enough that one can see the second inside the first.

So the same reaction that quietly draws down carbon dioxide is the principal non-marine deterioration mechanism in reinforced concrete worldwide, and the engineering response to it — denser mixes, thicker cover, longer service lives — is precisely a strategy for slowing the sink down. There is no version of this trade-off in which one side wins outright. A structure optimised to absorb carbon dioxide is a structure optimised to rust.

## Counted Going Out, Not Coming In

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The asymmetry in the accounting is the last oddity. The calcination emission appears in every national greenhouse gas inventory in the world, computed with the factor derived above. The carbonation sink appears in none of them: the 2006 IPCC Guidelines make no provision for it, and the question of whether and how to include it is still before the relevant expert bodies [17], [18]. It is, meanwhile, a routine term in the Global Carbon Budget, which means the same physical flux is inside the number climate scientists use and outside the number governments report.

It would be a mistake to read any of this as absolution. The sink is slow, it is capped at the process emissions alone, and it is largest in exactly the materials — thin renders, mortar beds, crushed demolition waste — that nobody builds for that reason. It cannot be banked in advance, and building more to capture more is self-defeating, since a tonne of new cement emits its carbon dioxide today and reclaims half of it over a century.

The stoichiometry does point somewhere useful, though, and it points backwards into the kiln rather than forwards into the wall. Alite, the fast-hardening phase cement is optimised for, is 73.7 per cent lime; belite, its slow sibling, is 65.1 per cent. Making clinker richer in the second lowers the limestone demand by about 12 per cent and the irreducible heat by 27 — 1,343 kJ per kilogram of phase against 1,848 — which is a larger saving than anything available downstream. The reason the industry has not simply done it is that belite takes months to develop strength rather than days, and construction schedules, not chemistry, are what set the composition of clinker.

What it does change is the arithmetic of what is already standing. The world has poured about 30 billion tonnes of concrete a year for a decade [19], and every square metre of it is slowly, silently running the kiln in reverse, at a rate fixed two centuries ago by a bricklayer's choice of raw material and by the diffusivity of carbon dioxide through a few millimetres of reacted paste.

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